

SUMMARY

PhD candidate in privacy-preserving ML with hands-on experience in ML, LLMs, and web engineering. Seeking a Data Science or ML internship to solve complex R&D challenges.

TECHNICAL SKILLS

**Programming:** Python, SQL, C/C++, Java, Bash  
**ML & Data Sci:** PyTorch, Scikit-learn, Hugging Face, OpenCV, Pandas, NumPy  
**Privacy ML:** Federated Learning, Differential Privacy (DP-SGD)  
**Databases & Tools:** Vector DBs, MySQL, SQLite, MongoDB, Docker, Git, CI/CD

EXPERIENCE

**PhD Researcher — Privacy-Preserving ML** Sep. 2025 – Present  
Eötvös Loránd University (ELTE) Budapest, Hungary

- Designing DP-SGD federated learning pipelines for synthetic data ( $\epsilon = 1$ ) and benchmarking privacy-utility trade-offs
- Developing the open-source IDaSec Practicum (8 labs on ML security & privacy) and mentoring graduate students
- Awarded the Stipendium Hungaricum Scholarship for academic excellence and research potential

**Software Engineer** Dec. 2024 – Mar. 2025  
RayaneBio Remote

- Architected and deployed a responsive e-commerce platform using React and Next.js, serving organic product customers
- Implemented secure user authentication and database models, improving user data management and transaction security
- Integrated customer feedback loops and testimonial features, elevating brand trust and increasing client engagement metrics

PROJECTS

**MyVellum** Dec. 2025 – Present  
LLM Orchestration, RAG, Docker, CI/CD

- Architected an LLM orchestration system with modular prompt templates for an AI-powered autobiography platform
- Developed vector-backed RAG pipelines with context-stitching, reducing hallucination rates in generated responses
- Integrated automated evaluation gates into CI/CD for relevance and safety metrics, ensuring quality across deployments
- Owned end-to-end API design and Docker deployment; mentored two engineers on prompt engineering best practices

**MS-AGHairNet** Jan. 2025 – Jun. 2025  
Python, PyTorch, OpenCV

- Designed a deep learning architecture for hair segmentation, achieving a 95.37% Dice coefficient on the benchmark dataset
- Engineered a data augmentation pipeline that improved generalization across image conditions, reducing overfitting by 15%
- Evaluated model performance with precision, recall, and F1-score; optimized inference pipeline for real-time deployment

EDUCATION

**PhD in Informatics — Privacy-Preserving Machine Learning** Sep. 2025 – Expected 2029  
Eötvös Loránd University (ELTE) Budapest, Hungary

Thesis: Synthetic data generation in artificial intelligence security (Supervised by Pr. Lendák Imre)

**M.Sc. in Artificial Intelligence** (GPA: 3.7/4.0) Sep. 2023 – Jun. 2025  
University of Khenchela Khenchela, Algeria

Thesis: Hair Dermoscopic Image Segmentation Using Deep Learning (Supervised by Pr. Dalal Bardou)

**B.Sc. in Mathematics and Computer Science** (GPA: 3.9/4.0) Sep. 2020 – Jun. 2023  
University of Khenchela Khenchela, Algeria

CERTIFICATIONS & TRAINING

- **Software Engineering Specialization** — ALX Africa Feb. 2025
- **AI Career Essentials (AiCE)** — ALX Africa Jul. 2024

LANGUAGES

Arabic (Native) | English (Professional Proficiency) | French (Intermediate) | Hungarian (Beginner)